



DEPLOYMENT TECHNIQUES AND STRUCTURAL ANALYSIS OF SOLAR SAILS FOR INTERPLANETARY MISSIONS

MINOR PROJECT END SEMESTER REPORT

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DECLARATION

We hereby certify that the work which is being presented entitled “Deployment strategies and structural analysis of solar sails for interplanetary missions” is in the partial fulfilment of the requirements for the minor project of the Bachelors of Technology in Aerospace Engineering and submitted in the Department of Aerospace Engineering the Punjab Engineering College, Chandigarh (Deemed to be University), is an authentic record of our own work carried out during the period from July 2023 to December 2023 under the supervision of Prof. Tushar Siag, Assistant Professor, Aerospace Department.

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Abstract

The development of solar sails is still in its early stages, but the technology has the potential to revolutionize space exploration. As the technology continues to develop, solar sails could become a major force in the future of space travel. The concept of solar sailing has been around for centuries, but it is only in recent decades that the technology has advanced to the point where it is feasible for real-world applications. This report presents a comprehensive analysis of the structural components and deployment techniques employed in solar sail propulsion modules, exploring their critical role in enhancing mission efficiency and ensuring successful space exploration. Solar sail technology harnesses the pressure of sunlight to propel spacecraft, offering a sustainable and cost-effective alternative to traditional propulsion methods. The structural analysis section investigates the materials, design, and engineering considerations that contribute to the development of robust solar sail structures. **Finite element analysis (FEA)** and computational simulations are employed to assess the structural integrity, thermal stability, and mechanical performance of the sail, ensuring its ability to withstand the harsh conditions of space travel. Furthermore, the report explores case studies of recent solar sail missions, highlighting successful deployment strategies and lessons learned from previous endeavors. These real-world examples provide valuable insights for future mission planning and contribute to the ongoing advancement of solar sail technology. By addressing key structural considerations and deployment challenges, this study contributes to the refinement of solar sail technology, paving the way for its broader integration into space exploration missions.

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1 Introduction

In the ever-evolving landscape of space exploration, propulsion technologies play a pivotal role in shaping the possibilities of deep-space missions. This project focuses on the revolutionary concept of solar sail propulsion modules, a promising avenue for sustainable and efficient propulsion in the vast expanse of outer space. Harnessing the power of sunlight to propel spacecraft, solar sails offer a compelling alternative to traditional propulsion methods, presenting an opportunity to redefine the future of space travel.

This project centers on two critical aspects essential for the success of solar sail missions: **structural analysis and deployment techniques**. The structural integrity of solar sail modules is paramount to withstand the myriad challenges of space travel, including radiation, and thermal extremes. Through advanced engineering analyses, we aim to explore and optimize the structural components, materials, and design considerations that contribute to the development of resilient solar sail structures.

Simultaneously, the project delves into the intricate realm of deployment techniques, a pivotal phase in the mission lifecycle. Efficient and reliable deployment mechanisms are crucial for realizing the full potential of solar sails, ensuring the timely and accurate unfurling of the sail membrane.

By amalgamating structural analysis and deployment techniques, this project strives to contribute to the cutting-edge advancements in solar sail technology. Real-world case studies from recent missions will be examined to distill valuable insights and lessons learned, providing a foundation for future mission planning and the refinement of solar sail systems.

Through this interdisciplinary investigation, we aspire to propel the field of space exploration forward, opening new avenues for sustainable and efficient propulsion technologies. The outcomes of this project are poised to inform space agencies, researchers, and engineers, ultimately advancing our collective understanding and capability to navigate the cosmos using the limitless energy of sunlight.

2 Literature Review

Solar Sails for Space Exploration– The Development and Demonstration of Critical Technologies in Partnership

M. Leipold, D. Kassing, M. Eiden and L. Herbeck

The paper discusses the diverse applications of solar sails in the solar system and cosmological exploration, including close circular polar orbits at solar distances ranging from 0.3 AU to 0.5 AU, Mercury orbiters, and sailcraft stationed sunward of the Sun-Earth Lagrangian point L1. The advantages of solar sails, such as lightweight design for launch, utilization of clean fuel, and continuous and inexhaustible propulsion, are highlighted. The proposed solar sail configuration features a square sail supported by diagonal booms, dividing it into four triangular segments. The sail structure comprises three main components: Carbon Fiber Reinforced Plastic (CFRP) booms, sail film segments, and a central deployment module. Challenges in sail fabrication involve using ultra-thin films and lightweight deployable booms capable of withstanding in-orbit loads, as well as efficiently packaging these elements into a small volume.

The technical hurdles outlined include the successful deployment of these lightweight structures in space and the control of the large but low-mass structure. Two major milestones are anticipated: the development of a 20 m x 20 m breadboard model of a fully deployable sail structure, co-funded by DLR (German Aerospace Centre) and ESA (European Space Agency), and cooperative pre-Phase-A studies with NASA/JPL, along with an ESA feasibility study, suggesting a low-cost technology-demonstration mission in Earth orbit to validate sail fabrication, packaging, storage, deployment, and control principles.

Two alternative mission routes are being studied: Route 1 (DAEDALUS): The sailcraft performs a number of orbit changes in Earth orbit to explore its potential for Earth-directed applications. The mission would end with a controlled reentry of the sailcraft after several months in orbit. Route 2 (ODISSEE): The sailcraft performs an orbit transfer to the Moon. A high-resolution camera on the sailcraft could explore lunar areas during a polar flyby or from lunar orbit.

The paper ends with a discussion on feasibility and a case for technology partnership with DLR and ESA.

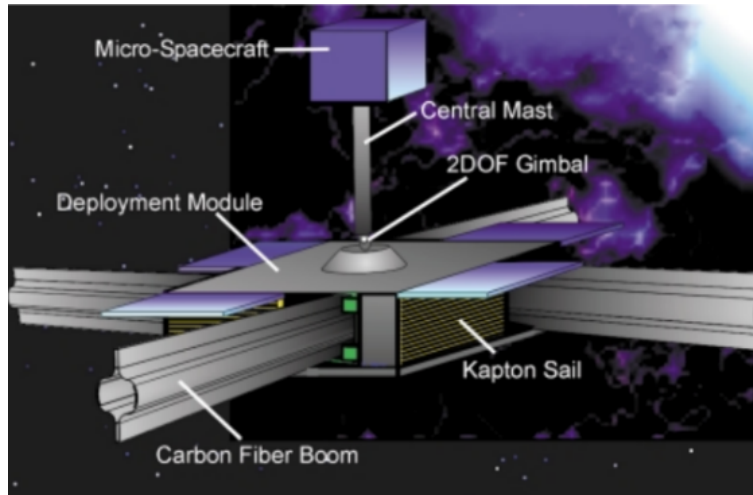


Figure 1: Central Hub Connections

ASSESSMENT OF ANALYSIS APPROACHES FOR SOLAR SAIL STRUCTURAL RESPONSE

Barmac K. Taleghani Army Research Laboratory, Vehicle Technology Directorate

David W. Sleight, Danniella M. Muheim, Keith Belvin, and John T. Wang NASA Langley Research Center

This article outlines an investigation into the vibration modes and deformations of solar sails using the MSC/NASTRAN and ABAQUS finite element programs. The primary objective was to establish a validated methodology for creating analytical models of solar sail structures to facilitate parametric design studies. Analyzing large solar sails requires considering geometrically nonlinear deformations, presenting additional computational challenges due to the relatively rigid booms and highly flexible sails.

The authors constructed component-level models for both the sail membrane and booms, utilizing NASTRAN and ABAQUS STATIC to identify beam and truss elements. Vibration frequencies (f_i) for the booms were computed under fixed-free end conditions. A model representing one quadrant of the 70 m solar sail membrane, featuring three corner cables, was created and assigned isotropic properties. The analysis employed the Newton-Raphson method to address nonlinearity.

The investigation included determining the maximum static deflection of the membrane, as well as calculating the vibration frequencies and mode shapes of the 70 m solar sail in its final nonlinear deformed state.

This paper describes the challenges associated with analyzing solar sails that feature ultra-

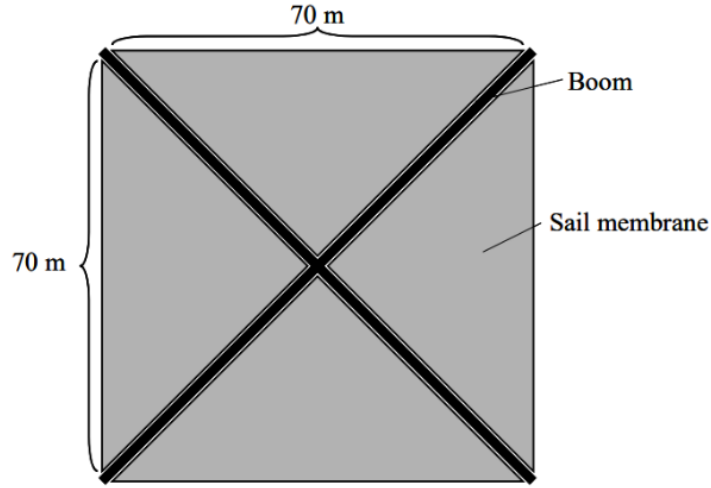


Fig. 1. Schematic of 70 m solar sail system

thin membranes and long, slender booms. The analysis difficulties stem from factors such as geometrically nonlinear effects and issues related to numerical convergence. The authors successfully addressed these challenges by employing NASTRAN and ABAQUS to analyze various configurations of solar sail components and assemblies.

To ensure the accuracy of their analyses, the researchers used closed-form solutions as a reference to validate the predictions made by finite element models for the response of boom components. Additionally, the finite element analyses of the sail component models were cross-validated against each other, confirming the reliability and consistency of the analytical results.

Finite Element Analysis about Solar Sail

Z Y Sheng, Jin Jin Zheng, H J Zhou , W H Huang , L G Sheng This research delves into the finite element analysis of a solar sail, a spacecraft propelled by sunlight pressure. The paper emphasizes the unique design of the solar sail model, highlighting structural challenges posed by its thin membranes and long booms. Using ANSYS for analysis, the study considers loads like solar radiation pressure and temperature, making specific assumptions for computational simplicity. Results suggest that deflection is primarily influenced by solar radiation pressure, with temperature exerting a minor impact. The conclusion asserts the model's credibility for a specific space position, offering insights for solar sail design and control applications.

The research carried out helps us identify the areas where assumptions can be taken for our mission. It is very difficult to carry out the analysis using real property of material and real environment in space. There are three reasons. First, the booms are very slender and long, and the films are ultra thin and large. Both of them are geometrical nonlinear. Second, there are many juncture that difficult to be defined in ANSYS. Third, the environment in space is complex; include solar radiation pressure, gravitation of stars and the effect of the change of temperature. In order to carry out a primary analysis and get an authentic result, some assumptions have been suggested.

- The material nonlinear of film is neglected, and treated as linear material.
- The extendibility of belt is neglected, but the rotation of belts can't be neglected.
- The films are treated as one material.
- Only the solar radiation pressure is considered, other planets' effects are all neglected. And the solar radiation pressure is considered as a constant.
- The temperature is considered as a constant.

3 Theory

Solar sail propulsion uses sunlight to propel vehicles through space by reflecting solar photons from a large, mirror-like sail made of a lightweight, reflective material. The continuous photonic pressure provides propellantless thrust to hover indefinitely at points in-space, or conduct orbital maneuver plane changes more efficiently than conventional chemical propulsion. Eventually, it could propel a space vehicle to tremendous speeds – theoretically much faster than any present-day propulsion system. Since the Sun supplies the necessary propulsive energy solar sails require no onboard propellant, the overall mass of the system may be reduced.

First generation sails will vary in size from 100 to 200 m depending on mission destination, and will typically be 3- axis stabilized. They will be compacted and stowed for launch. Once deployed, ultra-lightweight trusses will support the sails. Solar sails are composed of a flat smooth material, covered with a reflective coating, and supported by lightweight structures that are attached to a central hub. Near-term sails will likely use aluminized Mylar or CP1e. Both are proven materials previously flown in space. More robust sails might use a meshwork of interlocking carbon fibers.

The potential applications of solar sails are diverse and offer unique advantages in various space missions. Here’s a detailed exploration of the listed applications:

- **Special Orbit Dynamics:** Solar sails enable spacecraft to achieve highly non-Keplerian orbits with continuous low thrust, allowing for unique observational opportunities such as Earth observation and planetary science. Special orbits above the ecliptic can be utilized for space-based geoengineering and other scientific purposes.
- **Formation Flight of Solar Sails:** Solar sail formation flight is a novel concept that involves multiple solar sails working together in a coordinated manner. Formation flight can enhance the efficiency of deep space missions and contribute to missions like the GeoSail, where multiple sails on an inclined orbit may be more efficient than a single sail.
- **Active Space Debris Removal:** Solar sails can be employed for active space debris removal by using the sail as a drag sail to modify the orbits of debris. The Remove DEBRIS project has demonstrated the feasibility of using a drag sail for orbital transfer and debris removal.
- **Earth Polar Exploration:** Solar sails can be used to establish artificial balance above the ecliptic plane, allowing for Earth polar exploration. This can provide real-time

views of high latitude regions and poles for climate science, complementing traditional polar-orbit spacecraft limitations.

- **Exploration Mission of Extrasolar Planets:** Solar sails offer an alternative to conventional chemical propulsion for high-energy missions within the solar system. They can enable faster and more efficient exploration of planets, moons, and other celestial bodies, including those in the Kuiper Belt and beyond.
- **Flybys and Rendezvous of Small Celestial Bodies:** Solar sails, with their potential for unlimited Δv (velocity change), are well-suited for missions involving flybys, rendezvous, and sample return from small celestial bodies. Cost-effective exploration of multiple asteroids and comets is feasible using solar sailing spacecraft.
- **Interstellar Exploration:** Solar sails provide a potential solution for interstellar exploration by utilizing the continuous thrust from sunlight. Nano-spacecraft equipped with large sail areas and advanced membrane materials could achieve extremely high escape speeds, allowing for exploration beyond the solar system.

These applications demonstrate the versatility and efficiency of solar sails in addressing challenges and enabling missions that may be difficult or impractical with traditional spacecraft propulsion methods. As technology advances, solar sailing could play a significant role in the future of space exploration.

Basic components in solar sails are sail membranes, support booms, control vanes and a central hub. Flexible membranes exhibit complex mechanical behaviors, and their dynamics have always been difficult to predict. Current numerical simulation methods cannot effectively and quantitatively analyze motion and mechanical properties of membranes. Solar sail technology is an evolving field characterized by interdisciplinary design, currently in the stages of technological development and on-orbit demonstration. The present state of the art and technological advancements are assessed through the lens of solar sailing missions. It is evident that each specific technology does not comprehensively capture the current state of solar sail development. Furthermore, potential breakthroughs in various aspects, such as novel deployment methods and advanced material technologies, have the capacity to bring transformative advancements to solar sail technology.

Now some previous significant solar sailing missions are as follows:

3.1 IKAROS

The Interplanetary Kite-craft Accelerated by Radiation Of the Sun (IKAROS) satellite was built by Japan Aerospace Exploration Agency (JAXA), and was the first successful solar sail launched into space. Key features:

- A spinning solar sail
- Large LCD panels for attitude control to produce a center of mass (CoM) to center of pressure (CoP) offset
- They were used to control the spin of the satellite
- Changed the velocity to 400 meters per second

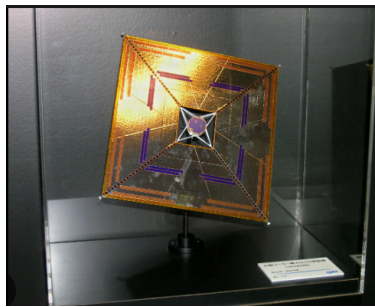


Figure 2: IKAROS Solar Sail

3.2 Nano sail

The NanoSail-D project was a 3U CubeSat-sized sail spacecraft. First successful in-orbit demonstration of a sail deployed by a CubeSat.

- A 10m² sail
- Semi-rigid booms
- A passive attitude control system which used permanent magnets to stabilize the satellite system.
- The NanoSail-D2 project did not make use of solar thrust and rather utilized aerodynamic drag to deorbit.
- The sail satellite had an orbit lifetime of eight months and re-entered the earth's atmosphere on 17 September 2011.

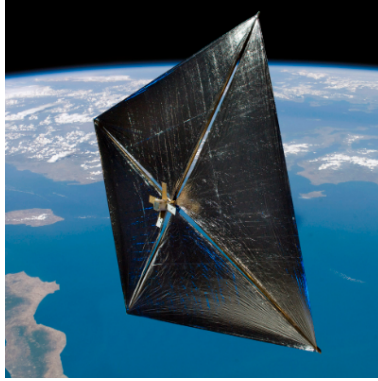


Figure 3: IKAROS Solar Sail

Other unsuccessful interplanetary solar sail missions:

- NEA Scout (Near-Earth Asteroid Scout): NEA Scout is a NASA mission launched in 2022 as part of the Artemis program. It aims to use solar sail propulsion to perform a flyby of a near-Earth asteroid.
- "Sunjammer" was a planned solar sail mission by NASA (1964) Sunjammer aimed to be one of the largest solar sail missions ever attempted and was intended to demonstrate the use of solar sail propulsion for deep-space exploration.

4 Deployment Strategies

The deployment of solar sails is a critical initial step essential for the proper functioning of the sail in various space missions. There are two primary categories of deployment methods:

- (1) Centrifugal Force-Based Deployment and
- (2) Boom-Based Deployment, with the deployment mechanism being a key component.

4.1 Spinning Solar Sails

In the case of spinning solar sails, such as the IKAROS model, a centrifugal force-based deployment mechanism is employed. This mechanism utilizes the rotation of the sail at speeds of up to 25 rpm, facilitated by centrifugal force, eliminating the need for additional supporting structures like booms. The spinning solar sail is characterized by a central hub as its primary structure. The central circular groove serves as the storage for the sail membrane, and four supporting cylinders extending from the central groove secure the membrane in place. During deployment, the sail membrane is wound around the hub rotator and extracted by the rotating rod as the hub undergoes rotation. The deployment process involves two stages: in the first stage, the sail is drawn out to form a cross shape, and in the second stage, the cross-shaped sail is released, achieving full deployment. This spinning deployment mechanism offers a distinct advantage due to its simplicity, lightweight design, and the absence of additional structural components like booms.

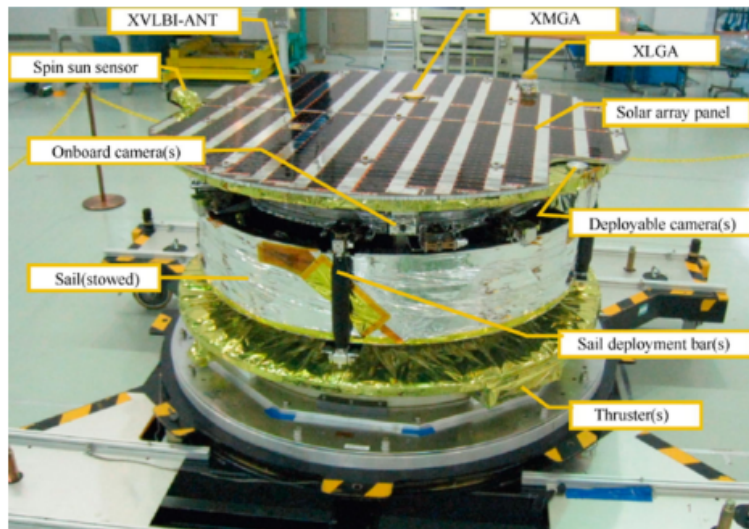


Figure 4: Spinning Solar Sail

4.2 Three Axis Stable Solar Sails

The deployment mechanism for three-axis stable solar sails is relatively intricate due to the nature of their deployment support. These sails are typically categorized into three types based on different driving methods: (1) Inflatable deployment; (2) Mechanical deployment; (3) Elastic deployment. Similar design concepts include components such as a central scroll, guide cylinders, motors, and composite booms. To prevent boom blossoming during deployment, a pinch device is commonly incorporated into the design.

For example, the Gossamer sail research team has devised two deployment mechanisms based on sail deployment booms. In one approach, rollers move radially inward as the boom extends from the central scroll. The other method utilizes bistable Carbon Fiber Reinforced Plastic (CFRP) booms. The latter employs 8 spring-loaded rollers to control Becu booms, while CFRP booms are inherently bistable, streamlining the deployment mechanism. Torsional springs are employed to prevent boom blossoming. Similarly, Fernandez et al. designed a deployable module featuring a fixed shaft connected to the satellite, around which four bistable booms are coiled. NASA has also proposed a Composite-Based Solar Sail System (CS3) for the NEA Scout Mission, aiming for deployment in the hundred square meter class.

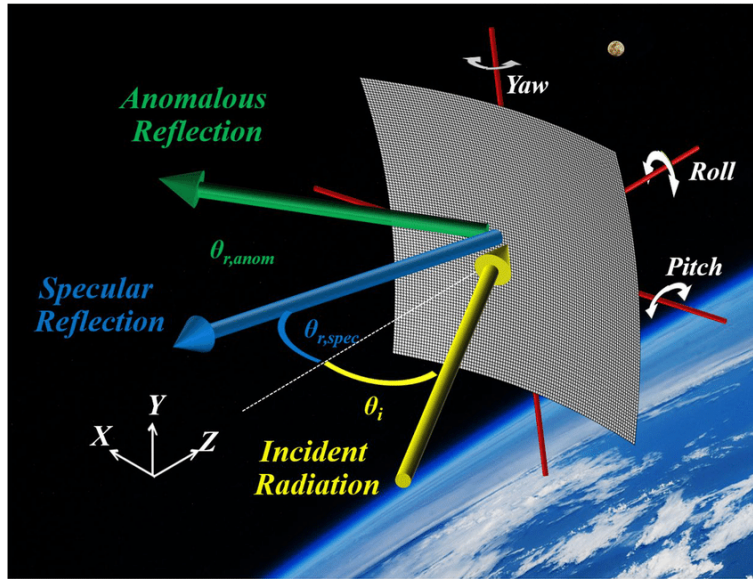


Figure 5: Three Axis Stable Solar Sail, Fig Ref: “Design and application of solar sailing: A review on key technologies”, by Pengyuan ZHAO , Chenchen WU , Yangmin LI

4.3 Heliogyro Structure

It utilizes angular momentum to eliminate the requirement of support that results in a lighter spacecraft. This kind of sail includes 12 spinning blades where each blade length is 4 miles long. So, the spacecraft moves toward Halley's Comet. The controlling of these blades can be done similarly to a helicopter by altering the pitch cycle to movement.

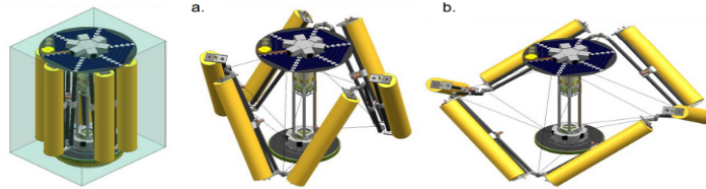


Figure 6: Heliogyro Solar Sail

5 Structural Analysis

In this study, a nonlinear static analysis, employing the Finite Element Method (FEM) with the ANSYS software, was conducted. The primary objectives were to determine membrane out-of-plane deflections caused by solar pressure, study the stress and strain distribution of structures, and analyze the strain energy distribution on the solar membrane. Structural responses, including out-of-plane deformations and wrinkles caused by solar pressure, were investigated. Numerical simulations explored the dependency of wrinkle amplitude on tension loads, considering a membrane truncated at the corners to avoid stress concentrations. While wrinkles' behavior was not the primary focus of this study, the structural response of a flat, square, thin-film membrane tensioned at the corners was examined.

5.1 Finite Element Model

The geometry of the solar sail was schematised in three main parts: the solar-sail membrane given by a square plate with 70 m of edge; the booms represented by four large wires; the central gub from which booms will be deployed. Booms were localised at 0.25 m from the membrane edges. The solar sail comprises four booms attached to a rigid central hub. Each individual boom has a length of 49.49 m.

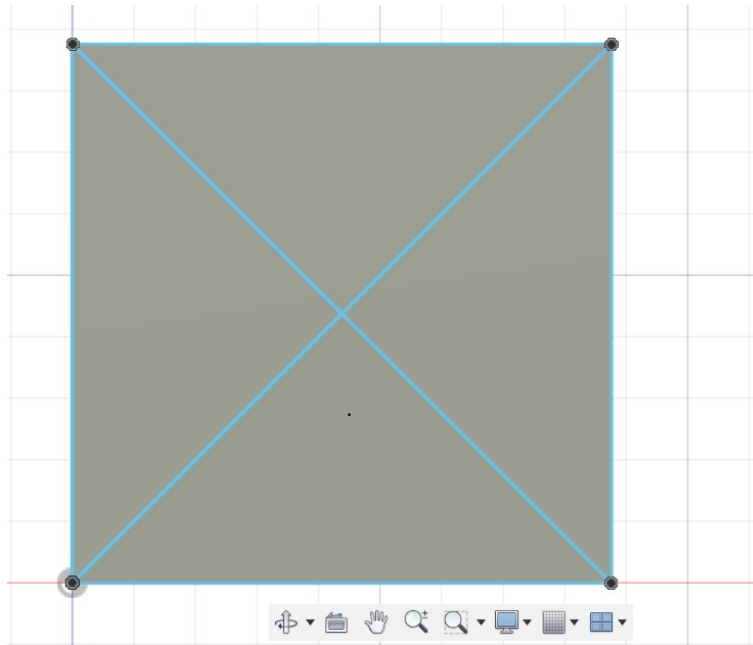


Figure 7: CAD of the solar sail

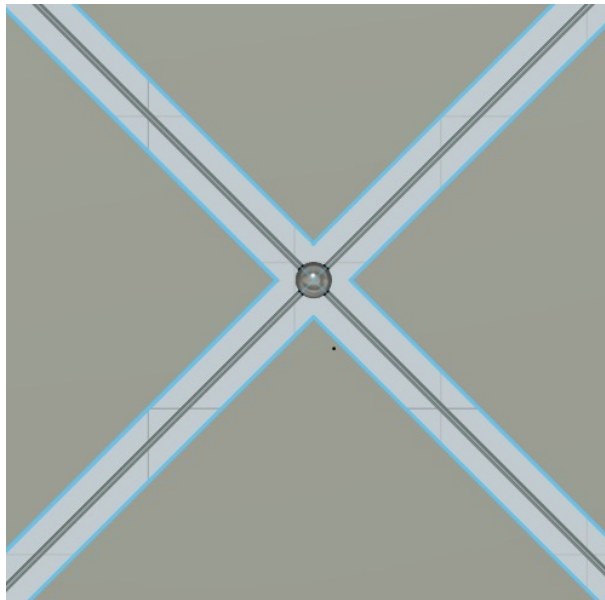


Figure 8: Central Hub of the Solar Sail

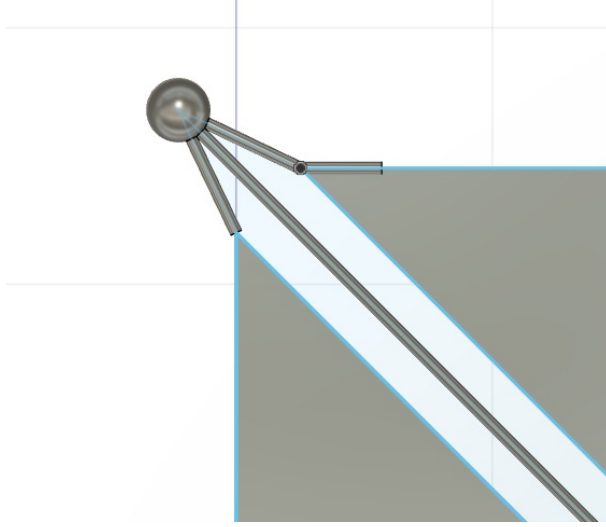


Figure 9: Booms attached to the sail membrane edges

The membrane flatness of the classical Xconfiguration is usually increased by dividing the entire sail into several strips. This helps the deployment and also reduces the stresses along the booms. On the other hand, both packaging and jointing of the membrane are challenging tasks and require particular attention. The solar-sail architecture presented in this work does not require the strip strategy described above. The material properties adopted in this work were taken from literature and are summarised in Table 1

Components	Materials	Thickness(m)	Modulus(N/m ²)	Poisson's ratio	Density (kg/m ³)
Booms	Composite	$2.54 \cdot 10^{-4}$	$131 \cdot 10^9$	0.30	2050
Cables	Kevlar	N/A	$62 \cdot 10^9$	0.36	1440
Membrane	Kapton	$7.62 \cdot 10^{-6}$	$2.48 \cdot 10^9$	0.34	1425.2

The finite element model was created through the discretization of the geometry, employing various element types. Specifically, the sail membrane was represented using membrane triangular elements (M3D3) with a constant thickness. However, to manage processing power limitations and meshing complexities, all other components, except for the sail membrane, were suppressed in the model.

5.1.1 Assumptions

Analyzing the structure in the real space environment poses significant challenges due to several factors. Firstly, the booms and films exhibit geometric nonlinearity, given their

slender and elongated nature, with films being ultra-thin and large. Furthermore, the films display material nonlinearity, while the booms exhibit state nonlinearity. Secondly, the intricate junctures in the model prove challenging to define accurately in ANSYS. Thirdly, the space environment introduces complexity, involving factors such as solar radiation pressure, gravitational effects from stars, and temperature variations. To facilitate a preliminary analysis and obtain credible results, certain assumptions have been introduced:

1. The material nonlinearity of the film is overlooked, and it is treated as a linear material.
2. The stretchability of the booms is disregarded, but the rotational movement of booms is considered crucial. Neglecting boom rotation can lead to convergence challenges, impacting the analysis.
3. Only solar radiation pressure is taken into consideration, with the effects of other planets being ignored. Additionally, solar radiation pressure is assumed to remain constant.
4. Temperature is assumed to be constant throughout the analysis.
5. The structure is assumed fixed in 5 points being central hub and the 4 boom connections.

5.1.2 The number of elements

To optimize the computational efficiency, the solar sail model is represented using **quadrilateral elements**. The elements are strategically distributed, with higher density in crucial regions like around orthogonal lines of sails, while maintaining lower density in secondary areas such as booms and other sail components. This approach yields a refined mesh with **390,680 nodes and 55,024 elements**, striking a balance between computational speed and result accuracy.

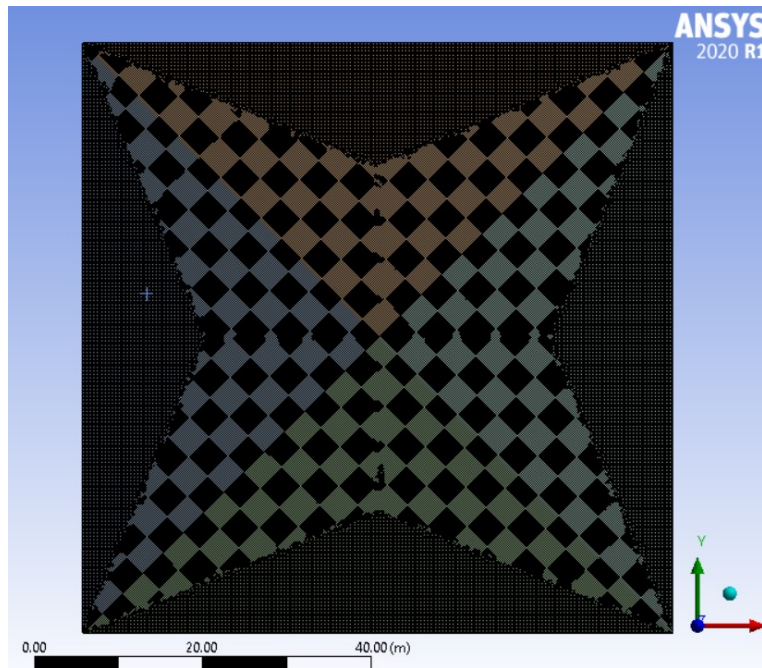


Figure 10: Meshing

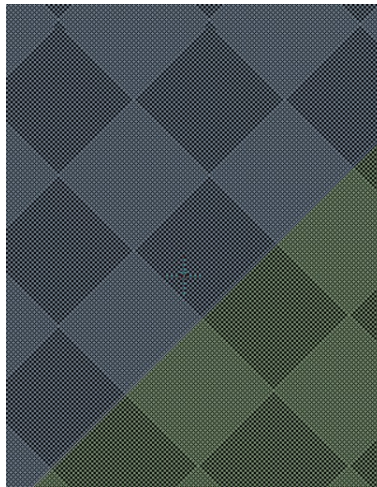


Figure 11: Closer view

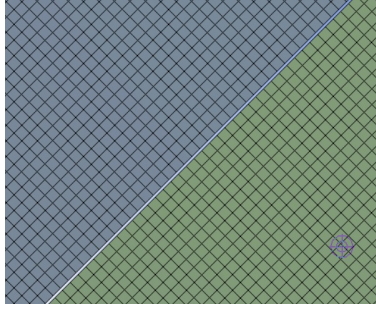


Figure 12: Structural Mesh Display

5.1.3 Load

The assumptions outlined earlier focus solely on accounting for solar radiation pressure and temperature. These assumptions encapsulate a representative characterization of the state of solar radiation pressure and temperature for the analysis.

5.1.4 Solar Radiation Pressure

From the figure above the below formula is deduced:

$$F = 2PA \cos \theta$$

To determine the load F , assumptions are made for the parameters P , A , and θ . The parameter P is contingent on the distance between the sun and the solar sail, with the chosen distance for analysis being 1 AU (the distance between the sun and Earth). Based on these parameters, the solar radiation pressure acting on the sail is calculated to be 9.12×10^{-6} Pa. It's noteworthy that the solar radiation pressure is maximal when θ (angle of incidence) is 0, and the solar sail must withstand this condition. Consequently, the analysis is deemed more reliable when θ is set to 0 compared to other directions.

5.2 Model Setup

Assuming perfect reflectivity of the sail membrane, the total pressure load resulting from Solar Radiation Pressure (SRP) is 9.12×10^{-6} N/m². This pressure was uniformly applied in the normal direction to the sail membrane. The applied boundary conditions on the solar sail involved fixing the positions of the five corners of the membrane and the central hub while allowing displacement along the four external edges and the four edges where the booms are positioned.

5.3 Results

5.3.1 Total Deformation

The external edges of the solar sail are likely to experience higher stress and deformation due to their position in the structure. The geometry of the sail, especially at its edges, can influence the distribution of forces and deformations. The external edges often bear the brunt of mechanical loading, including forces such as solar radiation pressure. As the sail intercepts sunlight, the resulting pressure and forces applied to the external edges contribute to The central portion of the external edges is often more susceptible to deformation because it represents a region where the sail material may have less structural rigidity compared to other parts. This can result in greater flexibility and deformation under external loads.

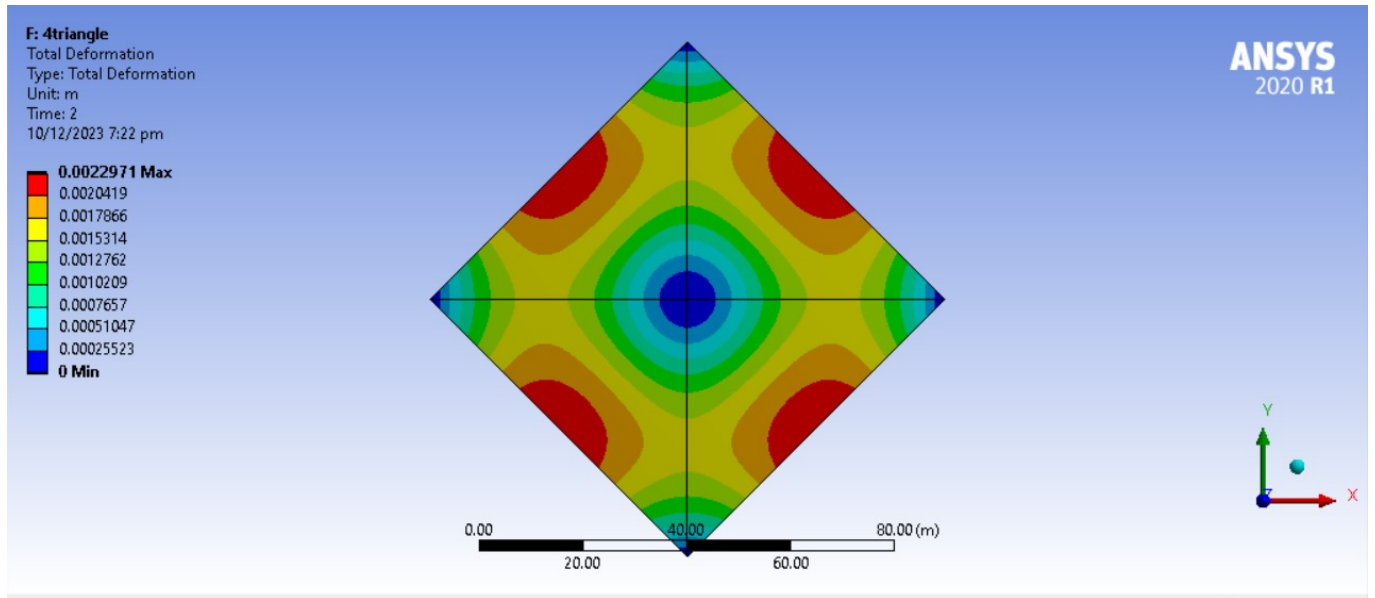


Figure 13: Total Deformation

5.3.2 Equivalent Stress

In the context of equivalent stress in a solar sail system, the solar radiation pressure, despite being extremely small and symmetrical, results in only a minimal increase in stress on the sail itself. The primary reason for this is that the load imposed by solar radiation pressure is effectively transferred to the booms. Consequently, while the stress increment on the sail is modest, the stress around and on the booms and central hub experiences a

substantial increase.

This distribution of stress highlights the significance of the structural elements in the solar sail system. The sail, being directly exposed to the ultra-small and symmetrical solar radiation pressure, undergoes a comparatively minor stress increase. However, the booms and central hub, which serve as crucial support structures, bear the brunt of the load, leading to a substantial rise in stress levels in these regions.

This observation underscores the importance of designing and reinforcing the booms and central hub to withstand the increased stress imposed by the solar radiation pressure.

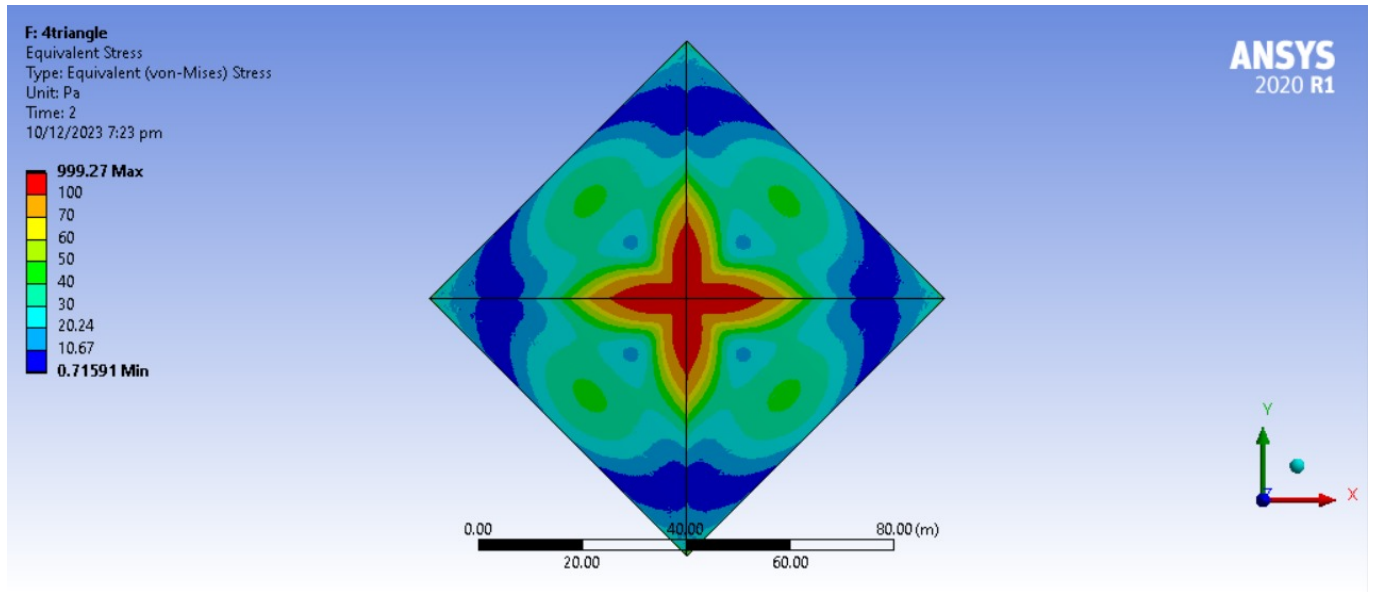


Figure 14: Equivalent Stress

5.3.3 Strain Energy

The contours of strain energy suggests that, within the solar sail system, the concentration of strain energy is notable at the fixed supports. This concentration occurs because the membrane surrounding these fixed supports experiences the maximum deformation.

The rationale behind this observation can be explained by the nature of fixed supports in a structural system. Fixed supports restrict movement and rotation, effectively anchoring the structure. When subjected to external forces, such as solar radiation pressure on the sail, the points around the fixed supports become critical zones where the membrane undergoes the most significant deformation.

As the membrane deforms, strain energy is stored within the material. The fixed supports

act as focal points where this strain energy is concentrated due to the restricted movement and deformation at these locations. This concentration of strain energy is indicative of the critical role played by fixed supports in absorbing and distributing the deformations within the solar sail system.

Understanding the distribution of strain energy is essential for optimizing the design and ensuring that the fixed supports can effectively manage the stresses and deformations induced by external forces.

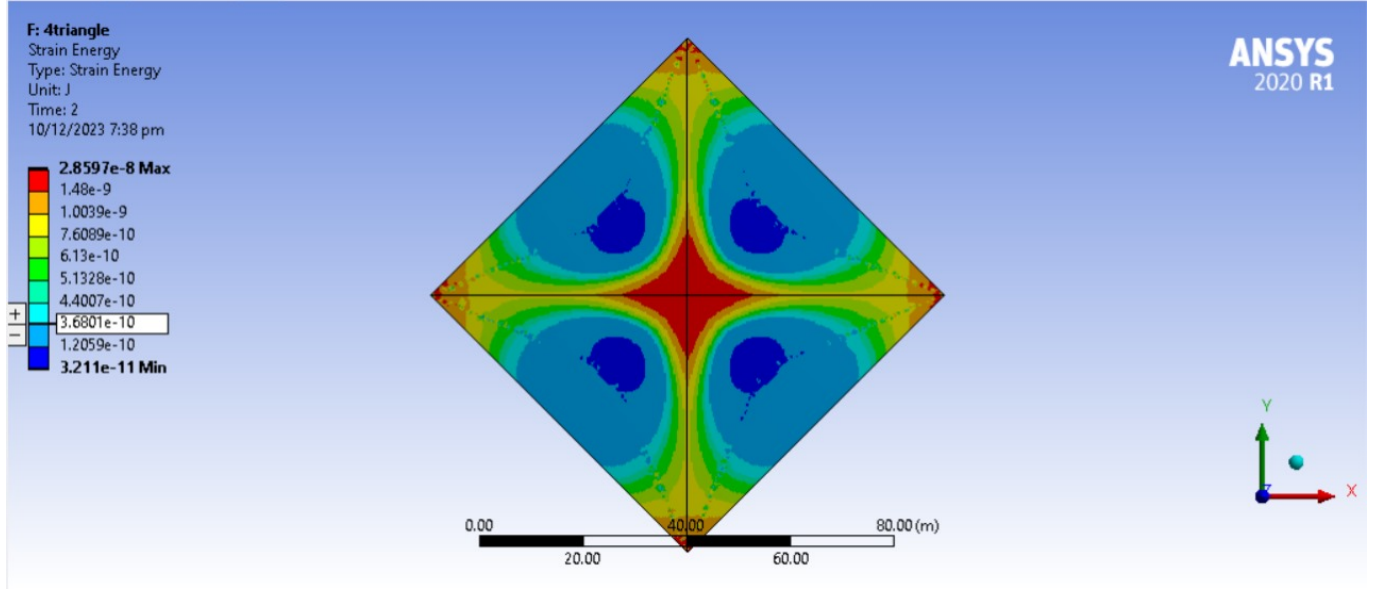


Figure 15: Strain Energy

5.3.4 Equivalent Elastic Energy

The observation regarding equivalent elastic strain in the solar sail system indicates that the central hub undergoes the highest level of deformation. As a result, the hub bears the brunt of the forces applied to the entire solar sail system. The design and configuration of the solar sail system may lead to a concentration of forces or stresses at the central hub. This can result in a higher level of elastic strain in comparison to other regions.

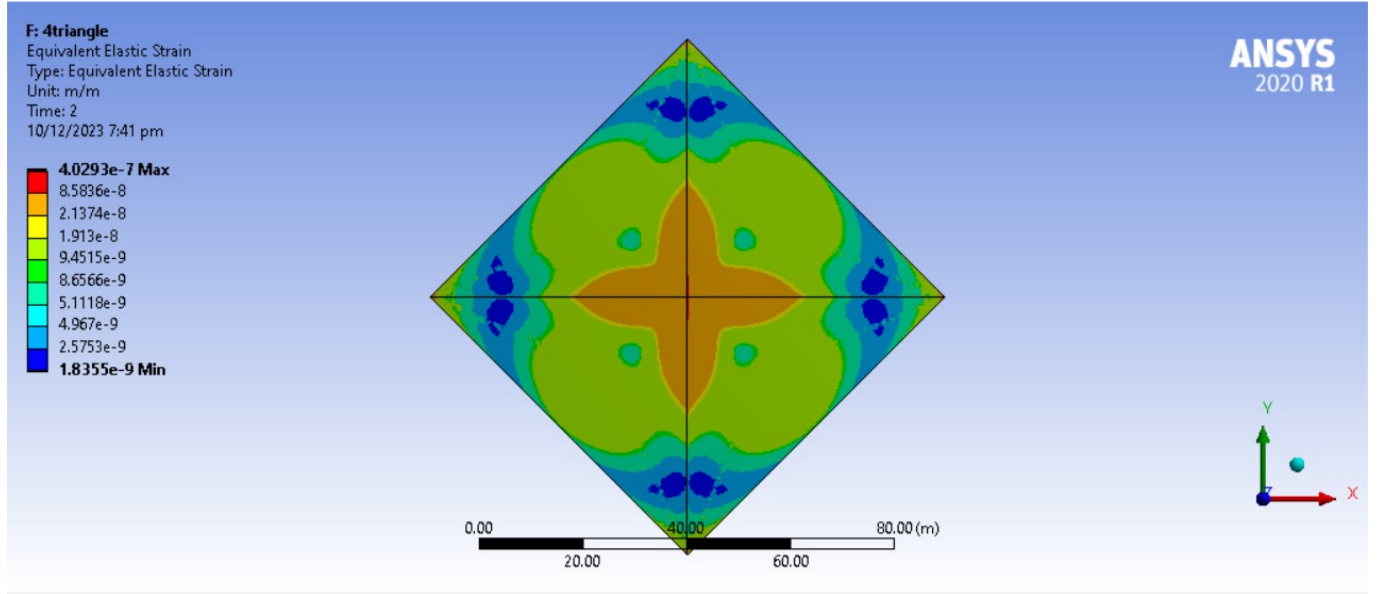


Figure 16: Equivalent Elastic Energy

5.4 Structural Analysis over a single membrane

To attain precise results in a triangular structure, a geometry slicing technique is employed. In this approach, a segment of the solar membrane is selected. Meshing triangular sections directly posed challenges due to their asymmetric nature. To overcome this, the triangular section is systematically divided into three parts. Subsequently, these individual sections are integrated into a single cohesive body, facilitating a more structured and manageable meshing process. The resultant mesh, demonstrating improved symmetry and ease of handling, is depicted in the figure below. This methodology ensures accurate representation and analysis of the triangular structure by addressing the complexities associated with its geometry during the meshing phase.

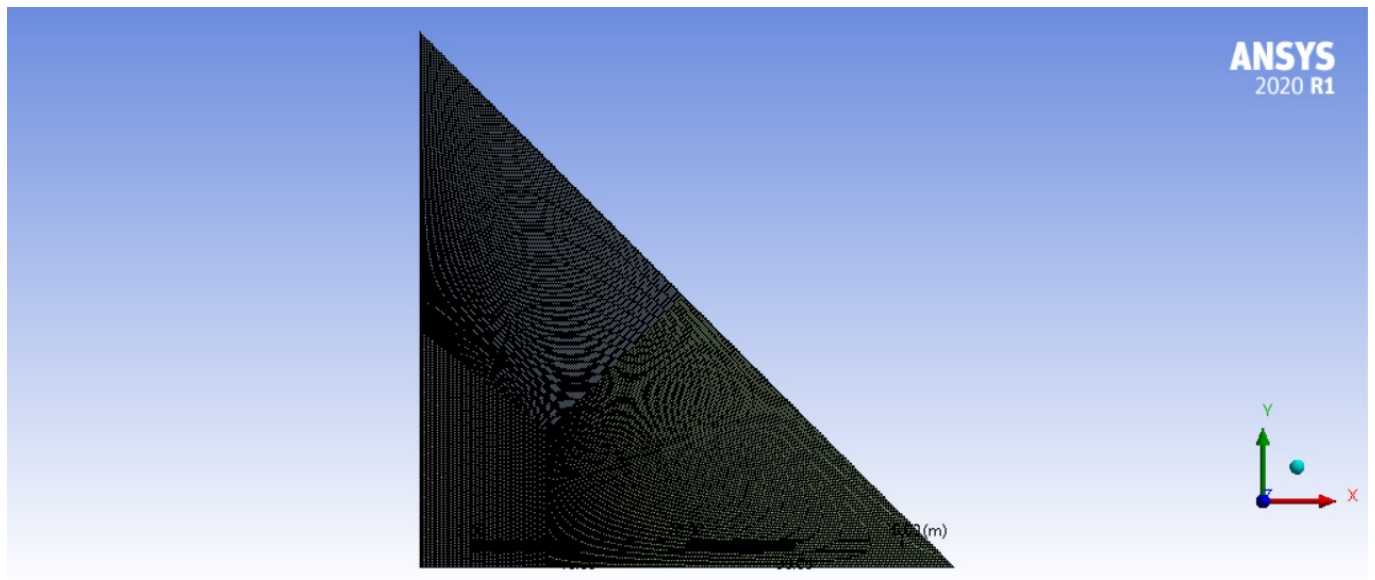


Figure 17: Meshing in triangle

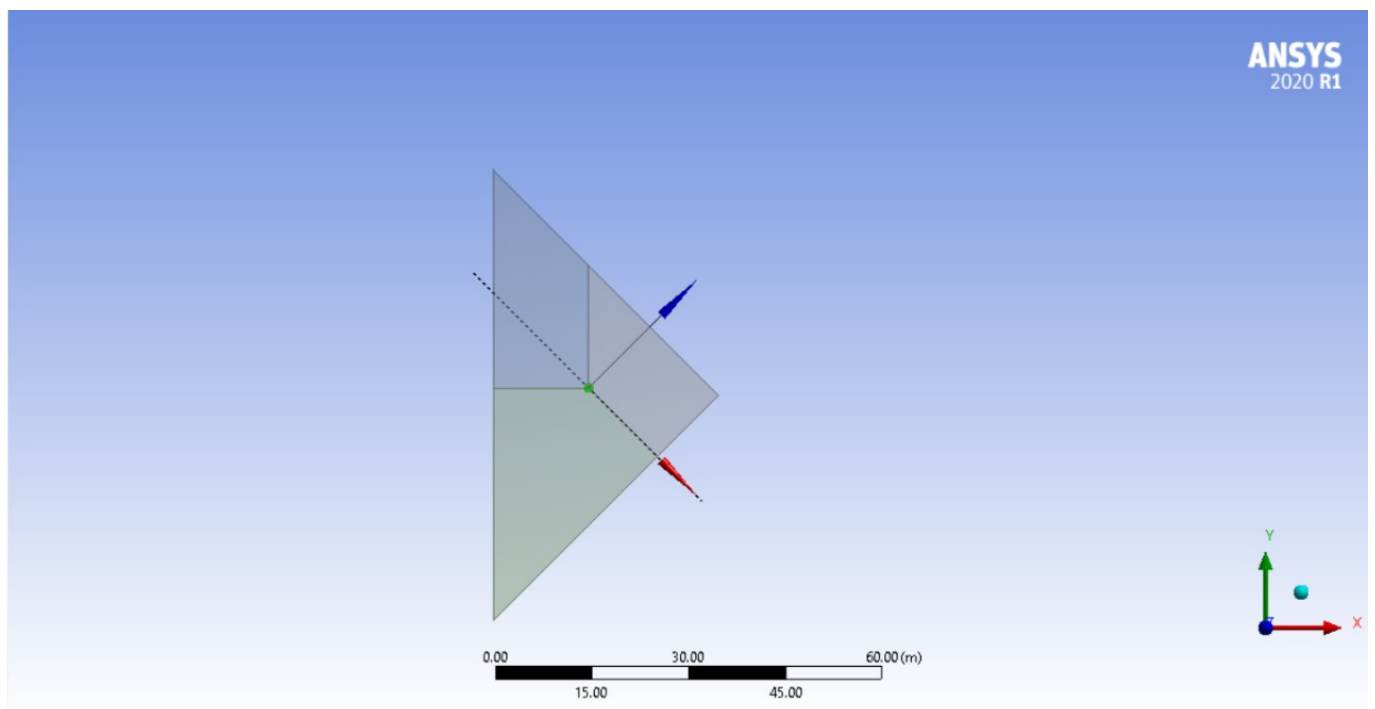


Figure 18: Close-up View

5.4.1 Numerical Results

In congruence with the characteristics observed in the fully deployed solar sail, the analysis indicates that the maximum deformation is concentrated at the center of the edges. This implies that the points equidistant from adjacent corners along the edges experience the most significant deformation. The prominence of deformation at these central edge locations suggests a critical area where structural responses to external forces or loads are particularly pronounced. Engineers and analysts would focus on understanding and mitigating deformations in these specific regions to enhance the overall performance and reliability of the solar sail during its deployment in space.

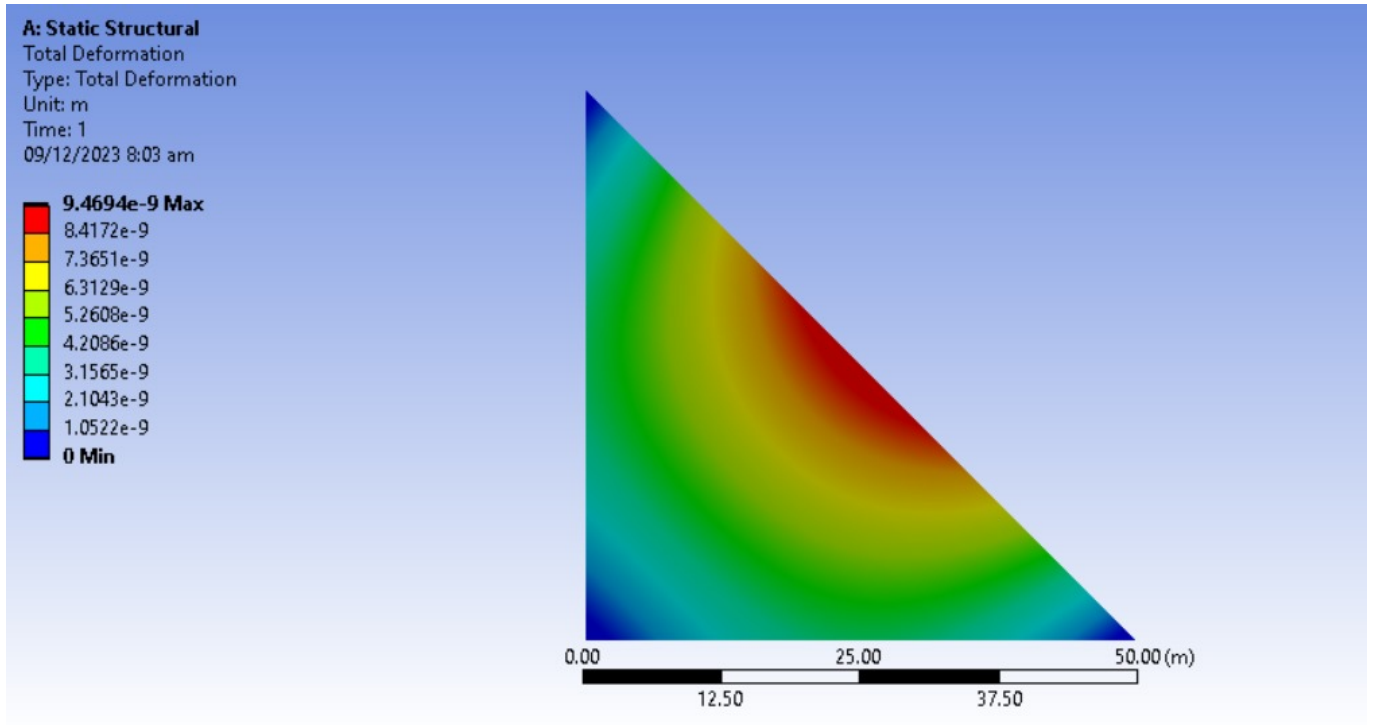


Figure 19: Total Deformation

5.5 Structural analysis of a partially deployed solar sails

We also conducted an analysis on a partially deployed geometry, as illustrated below. Meshing procedures were executed, and analogous boundary conditions and model setups were applied to the solar sail, maintaining consistency with the previous model configurations. This analysis allows for a comprehensive exploration of the structural behavior of the so-

lar sail under varying deployment conditions. Examining the response of the partially deployed geometry provides valuable insights into how the sail interacts with external forces in different states, aiding in the refinement of design considerations and structural optimizations for a broader range of operational scenarios.

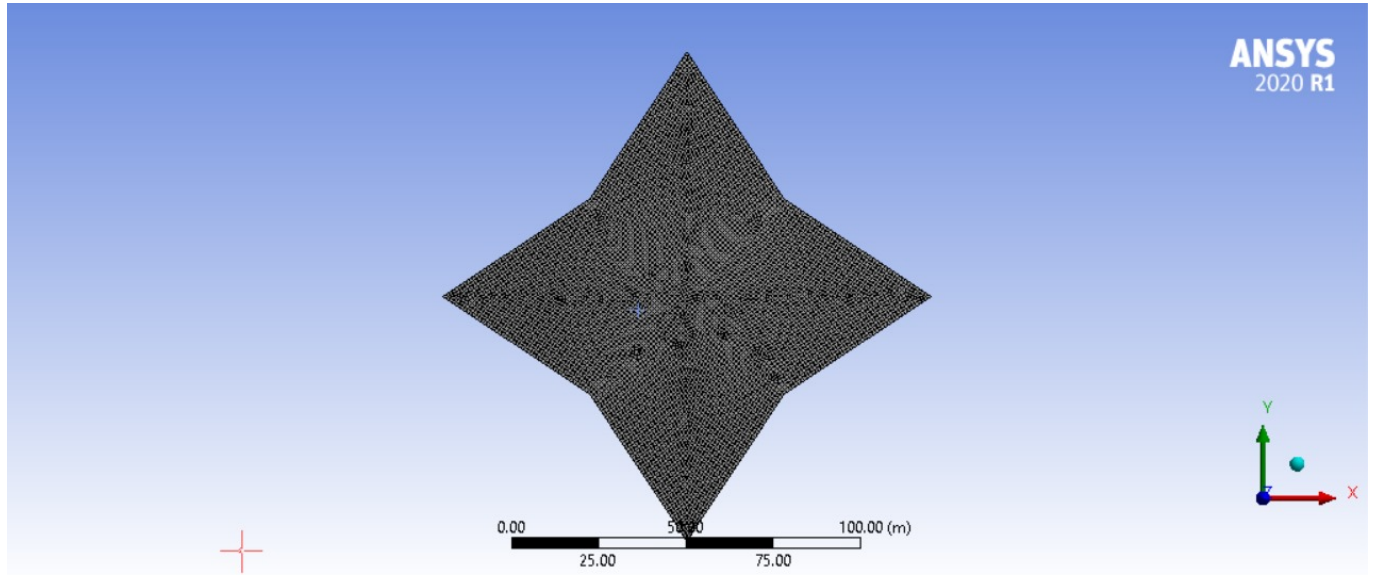


Figure 20: Meshing of deformed solar sail

5.5.1 Numerical Analysis

Total Deformation: The analysis of total deformation in the solar sail indicates that the maximum deformation occurs uniformly at all the outer edges of the membrane, while the minimum deformation is observed at the center. This deformation pattern highlights a consistent trend where the peripheral regions, being more exposed to external forces, experience heightened levels of deformation, while the central region remains relatively stable. Understanding this distribution is crucial for designing and optimizing the sail's structural integrity, ensuring it can effectively withstand external loads and maintain stability during deployment in space.

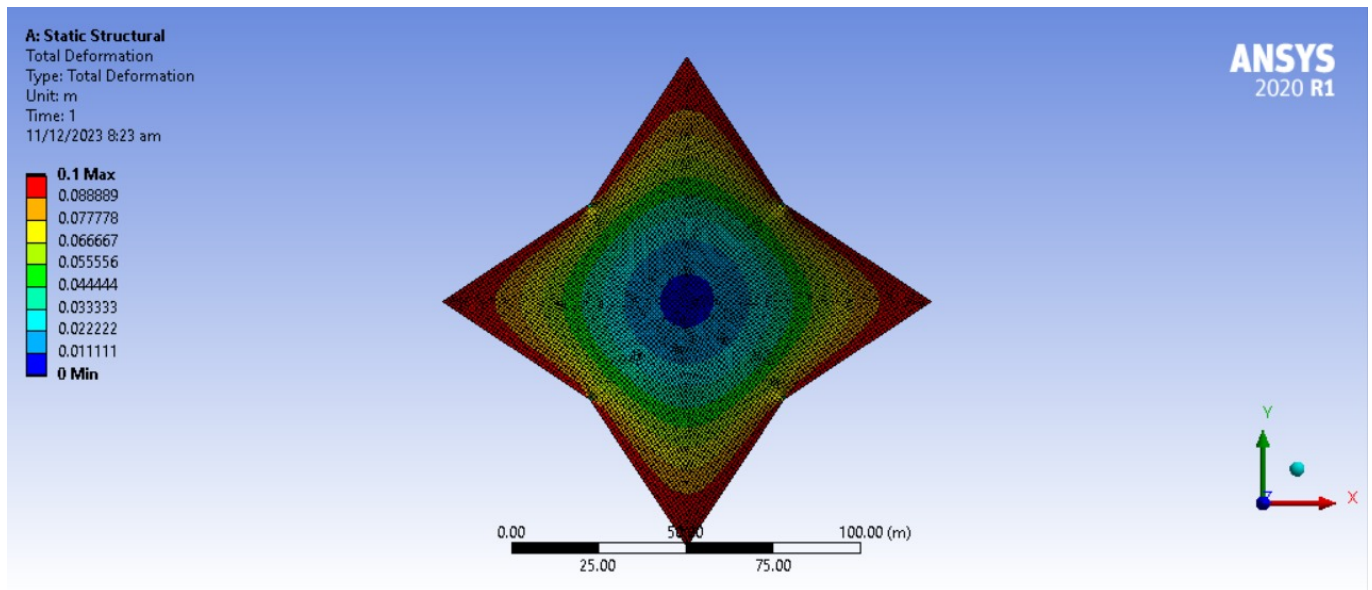


Figure 21: Total deformation

6 Challenges Faced in Solar Sail Analysis

1. Computational Limitations:

- The substantial size of the solar sail leads to an excessive number of elements.
- Personal computers struggle to compute within acceptable timeframes.

2. Trade-off Between Precision and Processing Time:

- To expedite the analysis, there's a need to reduce the number of elements.
- However, this reduction may compromise the precision of the results.

3. Thin Membrane and Booms:

- The membrane's thinness (7.62×10^{-6} m) complicates the meshing process.
- Booms, being even thinner, pose challenges in meshing and solving.

4. Pre-tension Application Difficulties:

- Applying pre-tension on the fine booms is impractical due to their delicate nature.

5. Simplified Environmental Conditions:

- Environmental conditions are simplified, neglecting all loads except solar radiation pressure.

These challenges highlight the intricacies involved in analyzing large-scale, delicate structures like solar sails, necessitating innovative approaches for accurate and computationally feasible simulations.

7 Scope and Future Work

The scope of structural analysis in the context of solar sails is crucial for designing robust spacecraft that can withstand the challenges of space. Structural analysis techniques, such as Finite Element Analysis (FEA) and dynamic analysis, play a pivotal role in simulating and optimizing the behavior of solar sail structures. This involves assessing how the sails respond to various loads, including radiation, micrometeoroid impacts, and deployment forces. The selection and analysis of materials become essential to ensure structural integrity and longevity in the harsh space environment. Additionally, understanding the deployment mechanisms and dynamics is vital for designing reliable systems. We can perform more advanced structural analysis with the use of more robust and powerful processing power. Meshing can be done with a higher number of nodes, and a better model made with the booms and connections designed more intricately.

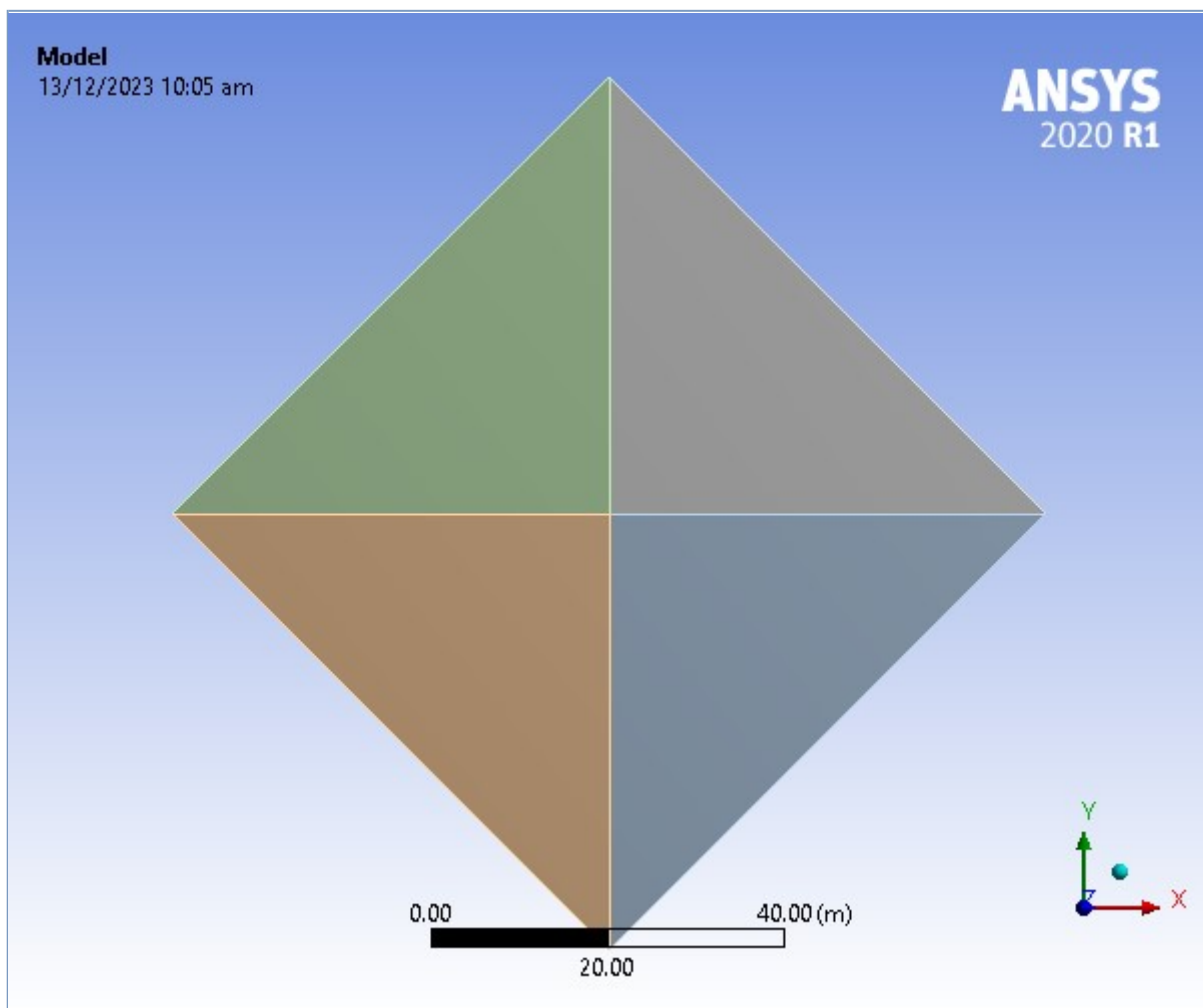
8 Conclusion

In the context of this project, the objective was to calculate the load exerted on a solar sail at a specific position in space. Subsequently, the solar sail, subjected to this calculated load, underwent a comprehensive analysis. The stress distribution was determined through the application of the finite element method, employing the ANSYS software. The obtained results were subjected to a comparative assessment against findings from other reference papers. The fact that the results aligned well with those from established references supports the conclusion that the outcomes of this analysis are deemed acceptable. This validation establishes the feasibility of the model used in this project. The significance of this approach lies in its potential applicability across various positions in space. By manipulating the parameters associated with solar pressure, the load at any given position in space can be precisely calculated. This flexibility allows for the systematic analysis of the solar sail under different conditions, showcasing its adaptability and robustness across varying scenarios. Overall, this project contributes to advancing our understanding of solar sail dynamics and reinforces the reliability of the model through successful comparisons with existing literature.



Project*

First Saved	Saturday, December 2, 2023
Last Saved	Sunday, December 10, 2023
Product Version	2020 R1
Save Project Before Solution	No
Save Project After Solution	No



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Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (F4)

Geometry

TABLE 2
Model (F4) > Geometry

Object Name	<i>Geometry</i>
State	Fully Defined
Definition	
Source	C:\Users\HP\Desktop\SS!!_files\dp0\SYS-4\DM\SYS-4.agdb
Type	DesignModeler
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	98.98 m
Length Y	98.98 m
Length Z	7.2e-003 m

Properties	
Volume	35.269 m ³
Mass	50259 kg
Scale Factor Value	1.
Statistics	
Bodies	4
Active Bodies	4
Nodes	390680
Elements	55024
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 3
Model (F4) > Geometry > Parts

Object Name	<i>Solid</i>	<i>Solid</i>	<i>Solid</i>	<i>Solid</i>
State	Meshed			
Graphics Properties				
Visible	Yes			
Transparency	1			
Definition				
Suppressed	No			
Stiffness Behavior	Flexible			
Coordinate System	Default Coordinate System			
Reference Temperature	By Environment			
Treatment	None			
Material				
Assignment	Kapton			
Nonlinear Effects	Yes			
Thermal Strain Effects	Yes			
Bounding Box				

Length X	49.49 m			
Length Y	49.49 m			
Length Z	7.2e-003 m			
Properties				
Volume	8.8173 m ³			
Mass	12565 kg			
Centroid X	16.497 m		-16.497 m	
Centroid Y	16.497 m	-16.497 m	16.497 m	-16.497 m
Centroid Z	3.6e-003 m			
Moment of Inertia Ip1	2.5645e+006 kg·m ²	8.5484e+005 kg·m ²	2.5645e+006 kg·m ²	2.5645e+006 kg·m ²
Moment of Inertia Ip2	8.5484e+005 kg·m ²	2.5645e+006 kg·m ²	8.5484e+005 kg·m ²	
Moment of Inertia Ip3	3.4194e+006 kg·m ²			
Statistics				
Nodes	97598	97771	97675	97636
Elements	13745	13769	13756	13754
Mesh Metric	None			

TABLE 4
Model (F4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	2
Material Assignments	0

Coordinate Systems

TABLE 5
Model (F4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. m
Origin Y	0. m
Origin Z	0. m
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]

Connections

TABLE 6
Model (F4) > Connections

Object Name	<i>Connections</i>
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	

Enabled	Yes
---------	-----

TABLE 7
Model (F4) > Connections > Contacts

Object Name	<i>Contacts</i>
State	Fully Defined
Definition	
Connection Type	Contact
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Auto Detection	
Tolerance Type	Slider
Tolerance Slider	0.
Tolerance Value	0.34995 m
Use Range	No
Face/Face	Yes
Face-Face Angle Tolerance	75. °
Face Overlap Tolerance	Off
Cylindrical Faces	Include
Face/Edge	No
Edge/Edge	No
Priority	Include All
Group By	Bodies
Search Across	Bodies
Statistics	
Connections	6
Active Connections	6

TABLE 8
Model (F4) > Connections > Contacts > Contact Regions

Object Name	Contact Region	Contact Region 2	Contact Region 3	Contact Region 4	Contact Region 5	Contact Region 6
State	Fully Defined					
Scope						
Scoping Method	Geometry Selection					
Contact	1 Face				2 Faces	
Target	1 Face				2 Faces	
Contact Bodies	Solid					
Target Bodies	Solid					
Protected	No					
Definition						
Type	Bonded					
Scope Mode	Automatic					
Behavior	Program Controlled					
Trim Contact	Program Controlled					
Trim Tolerance	0.34995 m					
Suppressed	No					
Advanced						
Formulation	Program Controlled					
Small Sliding	Program Controlled					
Detection Method	Program Controlled					
Penetration						

Tolerance	Program Controlled
Elastic Slip Tolerance	Program Controlled
Normal Stiffness	Program Controlled
Update Stiffness	Program Controlled
Pinball Region	Program Controlled
Geometric Modification	
Contact Geometry Correction	None
Target Geometry Correction	None

Mesh

TABLE 9
Model (F4) > Mesh

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Nonlinear Mechanical
Element Order	Program Controlled
Element Size	0.3 m
Sizing	
Growth Rate	Default (1.5)
Max Size	Default (0.6 m)
Mesh Defeaturing	Yes
Defeature Size	Default (1.5e-003 m)
Capture Curvature	Yes
Curvature Min Size	Default (3.e-003 m)
Curvature Normal Angle	Default (60.0°)
Capture Proximity	No
Bounding Box Diagonal	139.98 m
Average Surface Area	490.1 m ²
Minimum Edge Length	7.2e-003 m
Quality	
Check Mesh Quality	Yes, Errors
Target Skewness	Default (0.900000)
Target Jacobian Ratio (Corner Nodes)	Default (0.040000)
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled

Topology Checking	Yes
Pinch Tolerance	Default (2.7e-003 m)
Generate Pinch on Refresh	No
Statistics	
Nodes	390680
Elements	55024

TABLE 10
Model (F4) > Mesh > Mesh Controls

Object Name	Face Meshing	Body Sizing
State	Suppressed	
Scope		
Scoping Method	Geometry Selection	
Geometry	1 Face	4 Bodies
Definition		
Suppressed	Yes	
Active	No, Suppressed	
Mapped Mesh	Yes	
Constrain Boundary	No	
Type		Element Size
Element Size		0.2 m
Advanced		
Specified Sides	No Selection	
Specified Corners	No Selection	
Specified Ends	No Selection	
Defeature Size		Default (1.5e-003 m)
Behavior		Soft
Growth Rate		Default (1.5)
Capture Curvature		No
Capture Proximity		No

Static Structural (F5)

TABLE 11
Model (F4) > Analysis

Object Name	<i>Static Structural (F5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 12
Model (F4) > Static Structural (F5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	
Number Of Steps	2.
Current Step Number	1.

Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	On
Inertia Relief	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Unsymmetric
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Off
Output Controls	
Stress	Yes
Surface Stress	No
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
Contact Miscellaneous	No
General Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\HP\Desktop\SS!!_files\dp0\SYS-4\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes
Solver Units	Active System
Solver Unit System	mks

TABLE 13
Model (F4) > Static Structural (F5) > Analysis Settings
Step-Specific "Step Controls"

Step	Step End Time
1	1. s
2	2. s

TABLE 14
Model (F4) > Static Structural (F5) > Loads

Object Name	<i>Fixed Support</i>	<i>Pressure</i>	<i>Displacement</i>
State	Fully Defined		
Scope			
Scoping Method	Geometry Selection		
Geometry	5 Vertices	4 Faces	8 Edges
Definition			
Type	Fixed Support	Pressure	Displacement
Suppressed	No		
Define By		Normal To	Components
Applied By		Surface Effect	
Magnitude		9.12e-006 Pa (ramped)	
Coordinate System			Global Coordinate System
X Component			0. m (ramped)
Y Component			0. m (ramped)
Z Component			Free

FIGURE 1
Model (F4) > Static Structural (F5) > Pressure

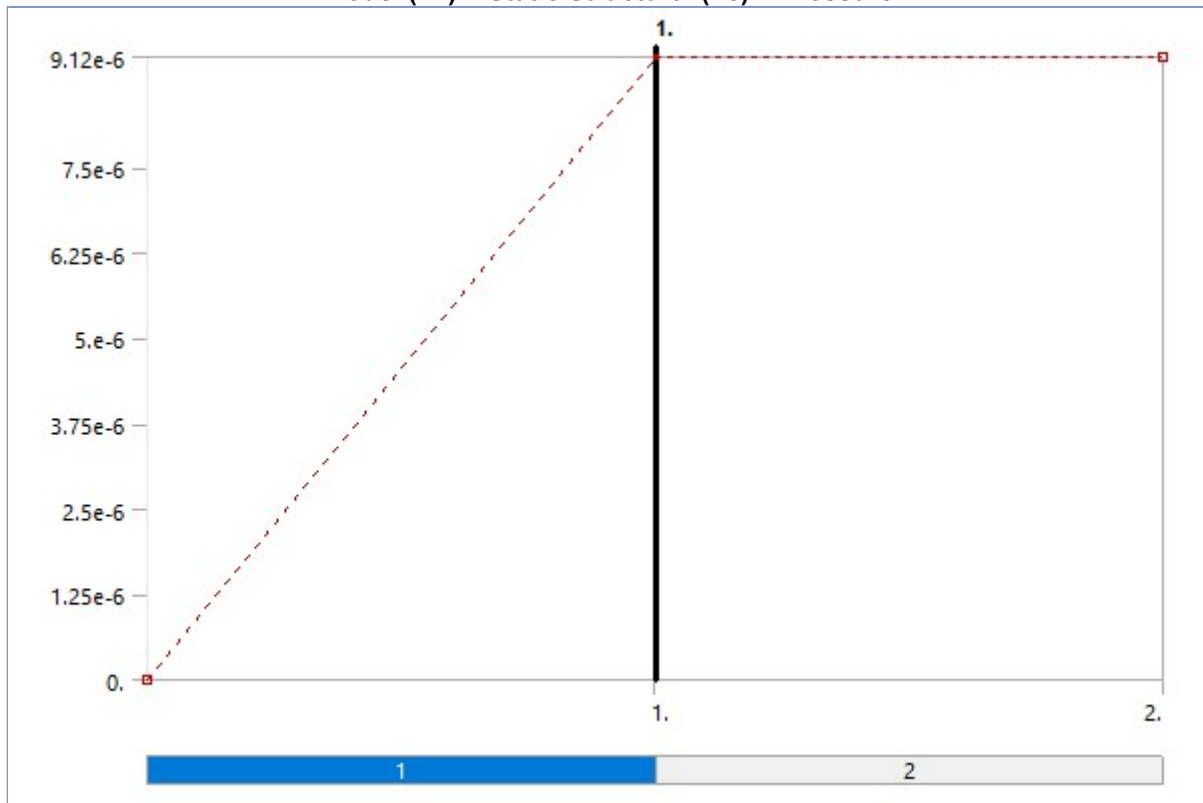


FIGURE 2
Model (F4) > Static Structural (F5) > Displacement



Solution (F6)

TABLE 15
Model (F4) > Static Structural (F5) > Solution

Object Name	<i>Solution (F6)</i>
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	18 m 3 s
MAPDL Memory Used	5.0303 GB
MAPDL Result File Size	142.25 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 16
Model (F4) > Static Structural (F5) > Solution (F6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All

FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 17
Model (F4) > Static Structural (F5) > Solution (F6) > Results

Object Name	Total Deformation	Equivalent Stress	Equivalent Elastic Strain	Strain Energy
State	Solved			
Scope				
Scoping Method	Geometry Selection			
Geometry	All Bodies			
Definition				
Type	Total Deformation	Equivalent (von-Mises) Stress	Equivalent Elastic Strain	Strain Energy
By	Time			
Display Time	Last			
Calculate Time History	Yes			
Identifier				
Suppressed	No			
Results				
Minimum	0. m	0.71591 Pa	1.8355e-009 m/m	3.211e-011 J
Maximum	2.2971e-003 m	999.27 Pa	4.0293e-007 m/m	2.8597e-008 J
Average	1.4753e-003 m	66.802 Pa	2.8301e-008 m/m	
Minimum Occurs On	Solid			
Maximum Occurs On	Solid			
Total				3.1944e-005 J
Minimum Value Over Time				
Minimum	0. m	0.71591 Pa	1.8355e-009 m/m	3.211e-011 J
Maximum	0. m	0.71591 Pa	1.8355e-009 m/m	3.211e-011 J
Maximum Value Over Time				
Minimum	2.2971e-003 m	999.27 Pa	4.0293e-007 m/m	2.8597e-008 J
Maximum	2.2971e-003 m	999.27 Pa	4.0293e-007 m/m	2.8597e-008 J
Information				
Time	2. s			
Load Step	2			
Substep	1			
Iteration Number	5			
Integration Point Results				
Display Option		Averaged		
Average Across Bodies		No		

FIGURE 3

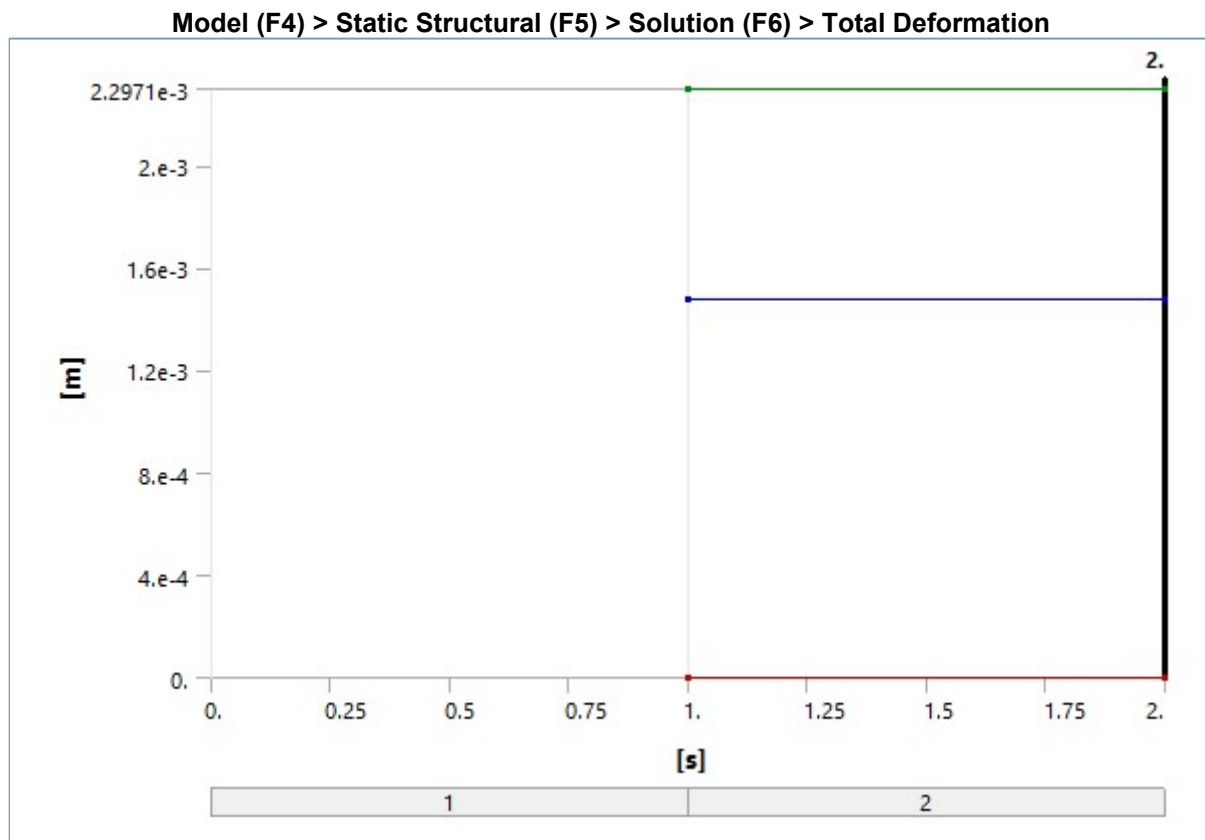


TABLE 18
Model (F4) > Static Structural (F5) > Solution (F6) > Total Deformation

Time [s]	Minimum [m]	Maximum [m]	Average [m]
1.	0.	2.2971e-003	1.4753e-003
2.			

FIGURE 4
Model (F4) > Static Structural (F5) > Solution (F6) > Equivalent Stress

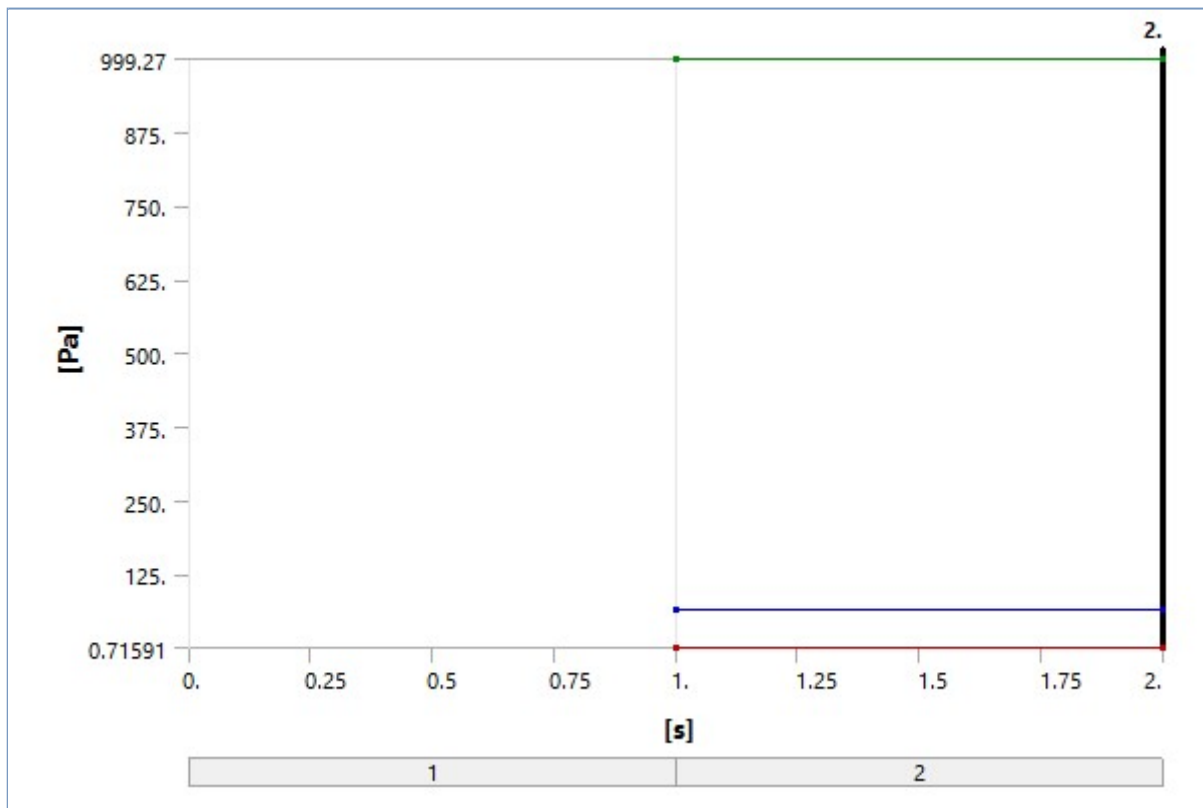
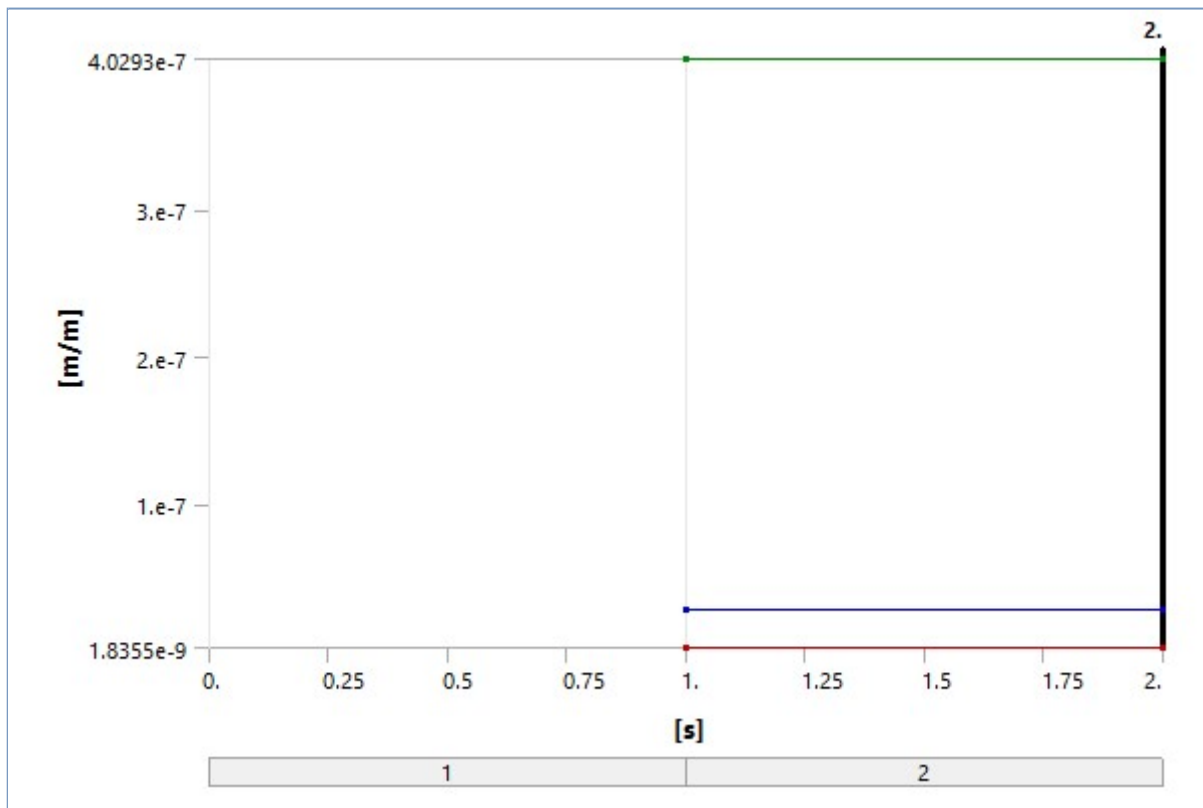


TABLE 19
Model (F4) > Static Structural (F5) > Solution (F6) > Equivalent Stress

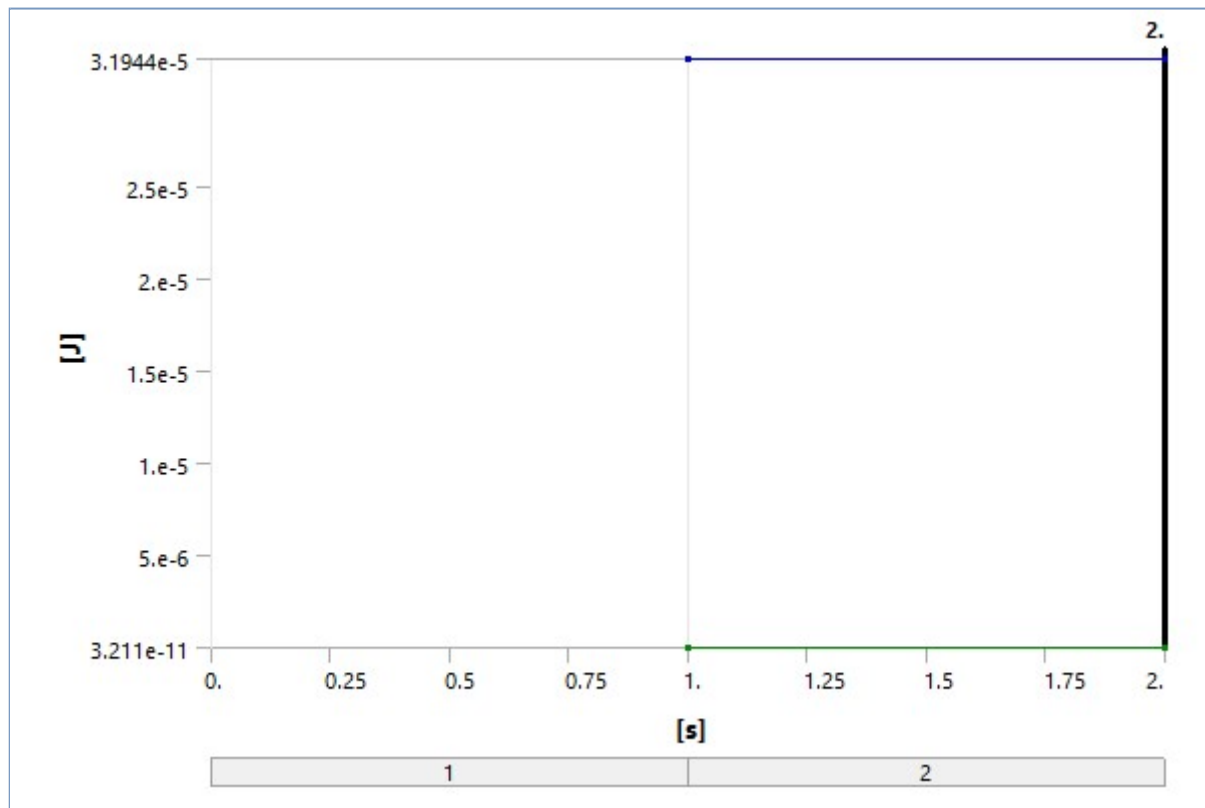
Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	0.71591	999.27	66.802
2.			

FIGURE 5
Model (F4) > Static Structural (F5) > Solution (F6) > Equivalent Elastic Strain

**TABLE 20****Model (F4) > Static Structural (F5) > Solution (F6) > Equivalent Elastic Strain**

Time [s]	Minimum [m/m]	Maximum [m/m]	Average [m/m]
1.	1.8355e-009	4.0293e-007	2.8301e-008
2.			

FIGURE 6**Model (F4) > Static Structural (F5) > Solution (F6) > Strain Energy**

**TABLE 21****Model (F4) > Static Structural (F5) > Solution (F6) > Strain Energy**

Time [s]	Minimum [J]	Maximum [J]	Total [J]
1.	3.211e-011	2.8597e-008	3.1944e-005
2.			

Material Data

Kapton

TABLE 22**Kapton > Constants**

Density	1425 kg m ⁻³
Coefficient of Thermal Expansion	2.3e-005 C ⁻¹
Specific Heat	875 J kg ⁻¹ C ⁻¹

TABLE 23**Kapton > Color**

Red	Green	Blue
138	104	46

TABLE 24**Kapton > Compressive Ultimate Strength**

Compressive Ultimate Strength Pa
0

TABLE 25**Kapton > Compressive Yield Strength**

Compressive Yield Strength Pa
2.8e+008

TABLE 26
Kapton > Tensile Yield Strength

Tensile Yield Strength Pa
2.8e+008

TABLE 27
Kapton > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
3.1e+008

TABLE 28
Kapton > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 29
Kapton > Isotropic Thermal Conductivity

Thermal Conductivity W m ⁻¹ C ⁻¹	Temperature C
114	-100
144	0
165	100
175	200

TABLE 30
Kapton > S-N Curve

Alternating Stress Pa	Cycles	R-Ratio
2.758e+008	1700	-1
2.413e+008	5000	-1
2.068e+008	34000	-1
1.724e+008	1.4e+005	-1
1.379e+008	8.e+005	-1
1.172e+008	2.4e+006	-1
8.963e+007	5.5e+007	-1
8.274e+007	1.e+008	-1
1.706e+008	50000	-0.5
1.396e+008	3.5e+005	-0.5
1.086e+008	3.7e+006	-0.5
8.791e+007	1.4e+007	-0.5
7.757e+007	5.e+007	-0.5
7.239e+007	1.e+008	-0.5
1.448e+008	50000	0
1.207e+008	1.9e+005	0
1.034e+008	1.3e+006	0
9.308e+007	4.4e+006	0
8.618e+007	1.2e+007	0
7.239e+007	1.e+008	0
7.412e+007	3.e+005	0.5
7.067e+007	1.5e+006	0.5
6.636e+007	1.2e+007	0.5
6.205e+007	1.e+008	0.5

TABLE 31
Kapton > Isotropic Resistivity

Resistivity ohm m	Temperature C
2.43e-008	0
2.67e-008	20
3.63e-008	100

TABLE 32
Kapton > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
2.48e+009	0.34	2.5833e+009	9.2537e+008	

TABLE 33
Kapton > Isotropic Relative Permeability

Relative Permeability
1

9 References

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- Solar Sailing Miranda Mesloh
- Principals of Solar Sailing - Richard Rieber
- Parametric Studies of Square Solar Sails Using Finite Element Analysis - David W. Sleight*
- ATTITUDE AND ORBITAL DYNAMICS MODELING FOR AN UNCONTROLLED SOLAR-SAIL EXPERIMENT IN LOW-EARTH ORBIT -Laura Pirovano(1), Patric Seefeldt(2), Bernd Dachwald(3), and Ron Noomen(4) and Danniella M. Muheim[†]
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